Credit Risk Portfolio Analytics — Architecture

Data flow: raw CSVs → ingestion → analytics → model → dashboard

Stage	Tool / library	Output
1. Raw data	5 CSVs in data/raw/ (pandas.read_csv)	customers, loan_accounts, bureau_credit, repayment_history, collections_events
2. Ingestion & cleaning	src/ingestion.py (pandas)	gender normalized, missing values imputed + flagged, referential-integrity checks logged
3. Feature engineering	src/features.py (pandas groupby/agg)	model_df (account grain) & customer_df (customer grain) — joins bureau/repayment/ collections rollups onto loans/customers
4. Analytics	src/analytics.py	default-rate cuts by dimension, cross-tabs, auto-generated insights
5. Model	src/model.py (scikit-learn LogisticRegression, XGBoost, SHAP)	AUC/KS/Gini, feature importance, SHAP drivers per account → artifacts/*.joblib, *.csv
6. Anomaly detection	src/anomaly.py (IsolationForest per peer segment + DPD-jump rule)	scored_accounts.parquet with anomaly_score, flag reasons
7. Ask-the-Portfolio (LLM)	src/llm_qa.py (OpenAI chat completions, rule-based fallback)	grounded Q&A answers
8. Dashboard	app/app.py (Streamlit, Plotly)	Portfolio view, Account view, Ask-the-Portfolio, Anomaly review

Where the LLM is called & what data is sent

- The LLM is called only inside src/llm_qa.answer_question(), invoked from the "Ask the Portfolio (AI)" page in app/app.py, one request per analyst question.
- What is sent: a small pre-aggregated JSON context (default-rate tables by dimension and cross-tab, model metrics, anomaly count) built by build_grounding_context() — never raw customer/account rows, names, or PII.
- The system prompt instructs the model to answer strictly from that JSON, quote exact figures, and decline if the question can't be answered from it.
- If OPENAI_API_KEY is not configured, the same grounded JSON is answered by a deterministic rule-based matcher (src/llm_qa._rule_based_answer) instead of an LLM call — the feature still works, with no risk of hallucination either way.

Both AI features embedded in the dashboard

- Ask the Portfolio: text box + answer on its own dashboard page, grounded in live-filtered data.
- Anomaly detection: review queue page listing flagged accounts with reasons, plus an analyst Confirm/Dismiss control that is persisted to artifacts/anomaly_decisions.csv (human-in-the-loop).